BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Objective Type) - Group - I

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Colour of iodine is:

(A) White
(B) Blue
(C) Purple
(D) Red

Q2. One of the following is a Lewis Base:

(A) NH_3
(B) BF_3
(C) H^+
(D) $AlCl_3$

Q3. The Conjugate acid of HPO_4^{2-} is:

(A) PO_4^{3-}
(B) $H_2PO_4^{2-}$
(C) $H_2PO_4^-$
(D) H_3PO_4

Q4. Main component of natural gas is:

(A) Methane
(B) Propane
(C) Butane
(D) Propyne

Q5. Oxidation of alkenes produce:

(A) Glyoxal
(B) Glycol
(C) Oxalic acid
(D) Formic acid

Q6. The organic compounds used as drugs to control bleeding are:

(A) Vitamins
(B) Proteins
(C) Lipids
(D) Glycerides

Q7. Almost 99% atmosphere's mass lies with in:

(A) 30 kilometer
(B) 35 kilometer
(C) 15 kilometer
(D) 11 kilometer

Q8. One of these pollutant is found in car exhaust fumes:

(A) CO_2
(B) O_3
(C) N_2
(D) H_2

Q9. Specific heat capacity of water is:

(A) 4.2 $kJg^{-1}K^{-1}$
(B) 4.2 $Jg^{-1}K^{-1}$
(C) 2.4 $kJg^{-1}K^{-1}$
(D) 2.4 $Jg^{-1}K^{-1}$

Q10. The disease that causes bone and tooth damage is:

(A) Fluorosis
(B) Hepatitis
(C) Cholera
(D) Jaundice

Q11. One of the following organic compound is found in gasoline:

(A) C_2H_4
(B) C_3H_8
(C) C_8H_{18}
(D) $C_{12}H_{26}$

Q12. Matte is mixture of:

(A) FeS and CuO
(B) Cu_2O and FeO
(C) Cu_2S and FeS
(D) CuS and FeO

Chemistry (Essay Type) (PART - I) - Group - I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Predict the direction of chemical reaction when Qc > Kc and Qc = Kc.

(ii) Define dynamic equilibrium and give an example.

(iii) What is ester functional group? Write its general formula.

(iv) Justify that organic compounds are used as food.

(v) Define the ketonic functional group with general formula.

(vi) What do you know about anthracite?

(vii) Describe the hydrogenation of ethene and ethyne.

(viii) How does combustion of methane take place?

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Why is a salt neutral? Explain with an example.

(ii) Name an acid used in the preservation of food.

(iii) How can you justify that Pb (OH) NO_3 is a basic salt?

(iv) How are proteins formed?

(v) How is gelatin obtained?

(vi) What are the effects of temporary hardness in water?

(vii) Mention disadvantages of detergents.

(viii) What is difference between biodegradable and non-biodegradable substances?

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) What is global warming?

(ii) Write two effects of acid rain.

(iii) Write down two serious effects of ozone depletion.

(iv) Give sources of CO and CO_2 (oxides of carbon).

(v) Write chemical formulae of copper glance and chalcopyrite.

(vi) How ammonical brine is carbonated? Give reaction.

(vii) Write a note on granulation of urea.

(viii) Write two uses of petroleum ether.

Chemistry (Essay Type) (PART-II) - Group - I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

Q5(a). (Marks 5) Explain the methods of removing permanent hardness.

Q5(b). (Marks 4) Explain reversible reaction with suitable example.

Q6(a). (Marks 5) What is petroleum? Describe any three fractions of petroleum.

Q6(b). (Marks 4) Describe any two chemical reactions of alkanes.

Q7(a). (Marks 5) Define salt. Explain with examples how soluble salts are prepared?

Q7(b). (Marks 4) Write a note on types of vitamins.

GROUP – II

Chemistry (Objective Type) - Group - II

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Acid rain effect the aquatic life by clogging fish gills because of:

(A) Lead metal
(B) Chromium metal
(C) Mercury metal
(D) Aluminium metal

Q2. In Solvay's process slaked lime is used to:

(A) Prepare CO_2
(B) Prepare quick lime
(C) Recover ammonia
(D) Form Na_2CO_3

Q3. Coal having 90% carbon content is called:

(A) Peat
(B) Lignite
(C) Anthracite
(D) Bituminous

Q4. Gas used for cleaning swimming pool:

(A) Iodine
(B) Chlorine
(C) Fluorine
(D) Bromine

Q5. Identify that which product is used as jet fuel:

(A) Kerosene oil
(B) Lubricating oil
(C) Diesel oil
(D) Fuel oil

Q6. At dynamic equilibrium:

(A) Reaction stops to proceed
(B) The amount of reactants and products are equal
(C) The speed of forward and reverse reaction are equal
(D) The reaction can no longer be reversed

Q7. One of the following is a reducing sugar:

(A) Glucose
(B) Maltose
(C) Sucrose
(D) Starch

Q8. One is a Lewis Base:

(A) NH_3 (B) BF_3

(C) H^+ (D) $AlCl_3$

Q9. Carbon monoxide is harmful to us because:

(A) It paralyses the lungs (B) It damages lung's tissue

(C) It reduces oxygen carrying ability of haemoglobin (D) It makes the blood coagulate

Q10. The order of reactivity of hydrogen halides with alkenes is:

(A) HI > HBr (B) HBr > HI

(C) HCl > HBr (D) HBr < HCl

Q11. Ion is the conjugate base of sulphuric acid:

(A) SO_3^{2-} (B) S^{2-}

(C) HSO_3^- (D) HSO_4^-

Q12. Vigorous reaction of water with — shatter (glass wear) into small pieces:

(A) Sodium (B) Caesium

(C) Strontium (D) Silicon

Chemistry (Essay Type) (PART - I) - Group - II

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) What do you mean by equilibrium constant?

(ii) What do you mean by extent of reaction?

(iii) Why melting and boiling points of organic compounds are low?

(iv) Define pitch and coke.

(v) Define catenation. Give an example.

(vi) Define isomerism with an example.

(vii) Give structure formula of isopentane and isobutane.

(viii) Why are the alkanes used as fuel?

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

(i) What is the function of DNA?

(ii) How plants synthesize carbohydrates?

(iii) Give a balanced chemical equation for Clark's method.

(iv) What is acute cadmium poisoning?

(v) Write a note on disease cholera.

(vi) Why is a salt neutral? Give an example.

(vii) Write the name of two acids used in manufacturing of fertilizers.

(viii) Define pH. What is pH of pure water?

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Describe the chemistry of removing the temporary hardness by boiling water.

(ii) What is the function of fertilizers?

(iii) State the method of swimming pool cleaning.

(iv) What is boiling point of water?

(v) Define ores.

(vi) What is function of pivot?

(vii) Name the raw materials used in Solvay process.

(viii) Write the formula of slaked lime.

Chemistry (Essay Type) (PART-II) - Group - II

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

Q5(a). (Marks 5) What is difference between soft water and hard water? Write disadvantages of hard water.

Q5(b). (Marks 4) Write macroscopic characteristics of dynamic equilibrium.

Q6(a). (Marks 5) What is the percentage of nitrogen in urea? Write down importance of urea.

Q6(b). (Marks 4) Write any four uses of ethene.

Q7(a). (Marks 5) Give sources of five different naturally occurring acids.

Q7(b). (Marks 4) Carbohydrates provide different usage to our body Explain.